MA1513 Exam (Examplify)

AY2023/2024 Semester II

MCQ/FIB (15 questions; 3 marks each)

1. Given a 3 x 4 linear system **Ax** = **b** with a general solution $\begin{cases} x_1 & = & s+t \\ x_2 & = & 3t-4s \\ x_3 & = & s+1 \\ x_4 & = & t-2 \end{cases}$. Which of the following is a particular solution of the homogeneous system **Ax** = **0**? (Pick all correct options)

   a. $\begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}$

   b. $\begin{pmatrix} 1 \\ -4 \\ 1 \\ -2 \end{pmatrix}$

   c. $\begin{pmatrix} 1 \\ 3 \\ 0 \\ 1 \end{pmatrix}$

   d. $\begin{pmatrix} 1 \\ 10 \\ 0 \\ 0 \end{pmatrix}$

   e. $\begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}$

The general solution of **Ax = b** in matrix form:

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = s\begin{pmatrix} 1 \\ -4 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 3 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \\ -2 \end{pmatrix}.$$

So the general solution of **Ax = 0** is: $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = s\begin{pmatrix} 1 \\ -4 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 3 \\ 0 \\ 1 \end{pmatrix}.$

By letting s = 0, t = 1, we get the particular solution $\begin{pmatrix} 1 \\ 3 \\ 0 \\ 1 \end{pmatrix}$, option (c).

By letting s = 1, t = 1, we get the particular solution $\begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}$, option (e).

The other 3 options are not linear combination $s\begin{pmatrix} 1 \\ -4 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 1 \\ 3 \\ 0 \\ 1 \end{pmatrix}.$

2. Suppose **A** = ($\mathbf{c_1}$ $\mathbf{c_2}$ $\mathbf{c_3}$) is a non-singular 3x3 matrix, where $\mathbf{c_1}$ **,** $\mathbf{c_2}$ **,** $\mathbf{c_3}$ are the three columns of **A**. Which of the following statements are true? (Pick all correct options)

   a. The three vectors $\mathbf{c_1}$ **,** $\mathbf{c_2}$ and $\mathbf{c_3}$ are coplanar in the XYZ-space.

   b. Any row echelon form of **A** has three pivot columns.

   c. The three rows of **A** form a basis for $\mathbf{R}^3$.

   d. **det A** is not zero.

   e. The non-homogenous system **Bx** = $\mathbf{c_3}$ is consistent where **B** is the 3x2 matrix ($\mathbf{c_1}$ $\mathbf{c_2}$).

Since **A** is non-singular, the columns of **A** is a linearly independent set. This means

a. The three vectors $\mathbf{c_1}$**,** $\mathbf{c_2}$ and $\mathbf{c_3}$ are NOT coplanar.

b. The three corresponding columns in the REF of **A** must be pivot columns.

c. **A** is non-singular also means the three rows of **A** is a linearly independent set, which form a basis for $\mathbf{R}^3$.

d. determinant of **A** is not zero.

e. $\mathbf{c_3}$ cannot be a linear combination of $\mathbf{c_1}$ and $\mathbf{c_2}$. The system **Bx** = $\mathbf{c_3}$ can be expressed in vector form as $x_1\mathbf{c_1} + x_2\mathbf{c_2} = \mathbf{c_3}$. Since there is no solution for x_1 and x_2, hence **Bx** = $\mathbf{c_3}$ must be inconsistent.

3. Which of the following have the same value as the 3x3 determinant $\begin{vmatrix} a & b & c \\ d & e & 0 \\ f & 0 & 0 \end{vmatrix}$ for any values of a, b, c, d, e and f? (Pick all correct options)

a. $\begin{vmatrix} f & 0 & 0 \\ d & e & 0 \\ a & b & c \end{vmatrix}$ = cef (multiply the diagonal of the triangular)

b. $\begin{vmatrix} 0 & 0 & f \\ 0 & e & d \\ c & b & a \end{vmatrix}$ = -cef (interchange row 1 and 3 to get triangular)

c. $\begin{vmatrix} -c & -b & -a \\ 0 & -e & -d \\ 0 & 0 & -f \end{vmatrix}$ = -cef (multiply the diagonal of the triangular)

d. $\begin{vmatrix} e & d & 0 \\ b & a & c \\ 0 & f & 0 \end{vmatrix}$ = -cef (inductive expansion along first row)

e. $\begin{vmatrix} 0 & 0 & f \\ c & b & a \\ 0 & e & d \end{vmatrix}$ = cef (interchange row 1 and 2, then row 2 and 3 to get triangular)

$\begin{vmatrix} a & b & c \\ d & e & 0 \\ f & 0 & 0 \end{vmatrix} = -cef.$

(Note that this matrix is not a triangular matrix.)

4. Which of the following vectors belong to the row space of $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 3 & 5 & 7 & 9 \end{pmatrix}$? (Pick all correct answers.)

a. (3, 6, 9, 12)
b. (4, 3, 2, 1)
c. (1, 2, 4, 8)
d. (3, 7, 5, 9)
e. (19, 29, 39, 49)

(a) is 3(1 2 3 4) , a scalar multiple of row 1.

(b) 5(1 1 1 1) – (1 2 3 4), a linear combination of row 1 and 2.

(e) 10(1 2 3 4) + 9(1 1 1 1), a linear combination of row 1 and 2.

(c) and (d) are not linear combinations of the three rows.

5. The set $\left\{\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right\}$ is a basis for the column space of which of the following matrices? (Pick all correct answers)

a. $\begin{pmatrix} 2 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}$

b. $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 3 \end{pmatrix}$

c. $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 3 \end{pmatrix}$

d. $\begin{pmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}$

e. $\begin{pmatrix} 1 & -1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}$

To answer this question, find the REF of each of the options and identify the pivot columns in each case. Those with more than 2 pivot columns will be ruled out, since we are looking for column space of dimension 2. (Option c has dimension 3, so is out.)

For options (a), (b) and (e), they each has two columns which form a basis for the column space. It is easy to check these columns are linear combinations of $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$, which can therefore be used as building blocks (basis) for their column spaces.

Option (d) is out, since there are columns in (d) that are not linear combinations of $\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

6. The projection **p** of the vector $\begin{pmatrix} 6 \\ 0 \\ 12 \end{pmatrix}$ onto the column space of the matrix $\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}$ is given by **p** = $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where a = 9 , b = 0 , c = 9 (FIB)

Find least squares solution $\boldsymbol{x}_0$ of the system $\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 0 \\ 12 \end{pmatrix}$, gives $\boldsymbol{x}_0 = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$.

The projection p = $\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} -3 \\ 6 \end{pmatrix} = \begin{pmatrix} 9 \\ 0 \\ 9 \end{pmatrix}$.

7. The coordinate vector of a vector **v** with respect to the basis S = { (1, 0, 0), (0, 1, 1), (0, -1, 1)} is given by $(\mathbf{v})_S$ = (2, 3, 5). Then the coordinate vector of **v** with respect to the basis T = { (1, -1, 0), (1, 1, 0), (0, 0, 1)} is given by $(\mathbf{v})_T$ = (a, b, c) where a = _2_, b = _0_, c = _8_ (FIB)

The vector **v** = 2(1, 0, 0) + 3(0, 1, 1) + 5(0, -1, 1) = (2, -2, 8)

Next we solve (2, -2, 8) = a(1, -1, 0) + b(1, 1, 0) + c(0, 0, 1) gives a = 2, b = 0, c = 8.

8. Which of the following subspaces of $\mathbf{R}^3$ are represented by the plane with equation 3x +y – 2z = 0 in the XYZ-space? (Pick all correct options)
   a. Span{ (3, 1, -2) }
   b. Row space of $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & -2 \end{pmatrix}$
   c. Nullspace of $\begin{pmatrix} 3 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$
   d. Column space of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & -3 & 2 \\ 3 & 0 & 1 \end{pmatrix}$
   e. Solution space of $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

A subspace represented by a plane should have dimension 2 and spanned by two vectors that lie on the plane.
Option (a) has dimension 1, (b) has dimension 3, and (e) has dimension 0. So these options are out.
Option (d) has dimension 2, and all the three columns satisfy the given equation of the plane. So this subspace is correct.

Option (c) says all vectors $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ that satisfy $\begin{pmatrix} 3 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ are in the nullspace. Expanding out the matrix equation gives 3x +y – 2z = 0, which is the same equation of the plane.

9. Let T: $\mathbf{R}^3 \to \mathbf{R}^3$ be a linear transformation such that $T\left(\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ and $T\left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

Then $T\left(\begin{pmatrix} 7 \\ 10 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where a = _8_, b = _10_, c = _3_ (FIB)

$$T\left(\begin{pmatrix} 7 \\ 10 \\ 4 \end{pmatrix}\right) = T\left(a\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + b\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = T\left(3\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + 4\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right) = 3\,T\left(\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}\right) + 4T\left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right)$$

$$= 3\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} + 4\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 8 \\ 10 \\ 3 \end{pmatrix}.$$

10. Given **A** = $\begin{pmatrix} 2 & 0 & 0 & 0 \\ * & 2 & 0 & 0 \\ * & * & 3 & 0 \\ * & * & * & 3 \end{pmatrix}$ is a diagonalizable matrix, where the * can be any random numbers.

Which of the following statements are correct? (Pick all correct answers)

a. The characteristic polynomial of **A** is $(x + 2)^2(x + 3)^2$

b. It is possible that one of the eigenspaces of **A** has dimension 1.

c. $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 3 \end{pmatrix}$ is an eigenvector of **A**

d. There is a matrix **P** such that $\mathbf{PAP}^{-1} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$

e. $\mathbf{A}^n$ is a diagonalizable matrix for any positive integer n.

(a) The characteristic polynomial should be $(x - 2)^2(x - 3)^2$ instead.
(b) Since **A** is diagonalizable, dimensions of both eigenspaces must be equal to the multiplicities (the powers in the characteristic polynomial), which are both 2.
(c) Just multiply A to the vector in (c).
(d) Since A is diagonalizable with eigenvalues 2, 2, 3, 3, there is a matrix **Q** such that $\mathbf{Q}^{-1}\mathbf{AQ} = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$. Then $\mathbf{P} = \mathbf{Q}^{-1}$ will satisfy the equation in (d).
(e) Since **A** is diagonalizable, any power of **A** is also diagonalizable. For example, using the **Q** in (d) above, $\mathbf{Q}^{-1}\mathbf{A}^n\mathbf{Q} = \begin{pmatrix} 3^n & 0 & 0 & 0 \\ 0 & 3^n & 0 & 0 \\ 0 & 0 & 2^n & 0 \\ 0 & 0 & 0 & 2^n \end{pmatrix}$.

11. Given a 2x2 matrix **A** with nullspace given by span$\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$ and $\mathbf{A}^3\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \end{pmatrix}$. Then **A** is given by $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ where a = _-2_, b = _2_, c = _-4_, d = _4_ (FIB).

Nullspace of **A** is the same as eigenspace of **A** with eigenvalue 0. So we know **A** has eigenvalue 0 with eigenvector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$\mathbf{A}^3\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \end{pmatrix} = 8\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, which means $\mathbf{A}^3$ has eigenvalue 8 with eigenvector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$. So we deduce that **A** has eigenvalue 2 with same eigenvector.

Let $\mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$. We can solve for $\mathbf{A} = \mathbf{PDP}^{-1} = \begin{pmatrix} -2 & 2 \\ -4 & 4 \end{pmatrix}$.

12. Given the SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$, which of the followings are solutions of the SDE? (Pick all correct answers)

a. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} \sin 2t \\ e^{3t} \cos 2t \end{pmatrix}$

b. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} \sin 2t \\ -e^{3t} \cos 2t \end{pmatrix}$

c. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} \cos 2t \\ e^{3t} \sin 2t \end{pmatrix}$

d. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} -e^{3t} \cos 2t \\ e^{3t} \sin 2t \end{pmatrix}$

e. $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} e^{3t} (\cos 2t + \sin 2t) \\ e^{3t} (\cos 2t - \sin 2t) \end{pmatrix}$

The matrix has a complex eigenvector 3 + 2i with eigenvector $\begin{pmatrix} i \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + i\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

So we have a = 3, b = 2, **p** = $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and **q** = $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Fundamental set of solutions: $X_{Re} = e^{3t}\begin{pmatrix} -\sin 2t \\ \cos 2t \end{pmatrix}$ and $X_{Im} = e^{3t}\begin{pmatrix} \cos 2t \\ \sin 2t \end{pmatrix}$

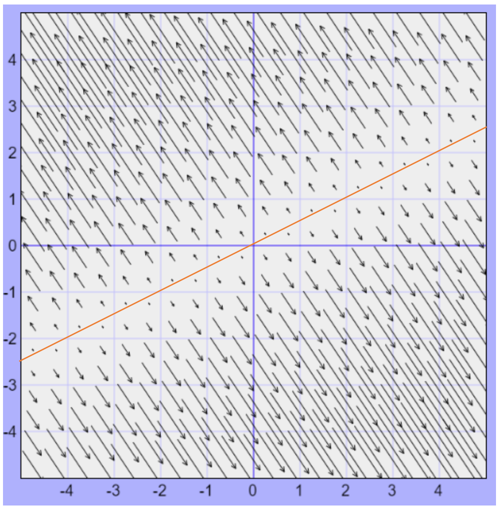
General solution: $c_1 e^{3t}\begin{pmatrix} -\sin 2t \\ \cos 2t \end{pmatrix} + c_2 e^{3t}\begin{pmatrix} \cos 2t \\ \sin 2t \end{pmatrix}$.

Option (b): Take $c_1 = -1, c_2 = 0$.

Option (c): Take $c_1 = 0, c_2 = 1$.

No solution for c_1 and c_2 for all the other options.

13. The direction field of a 2 by 2 SDE **y**′ = **Ay** is given below.



From the phase portrait, we deduce **A** has an eigenvalue 0 with eigenvector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$, and another positive eigenvalue r with eigenvector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$, since all the little arrows of the direction field are pointing in that direction. This is a special case (not a star node), with infinitely many equilibrium points along the red line. Since 0 is an eigenvalue, **A** is singular.

Which of the following statements are correct? (Pick all correct answers)

a. The equilibrium point of the SDE is a star node

b. There are infinitely many equilibrium points for the SDE

c. **A** has only 1 linearly independent eigenvector

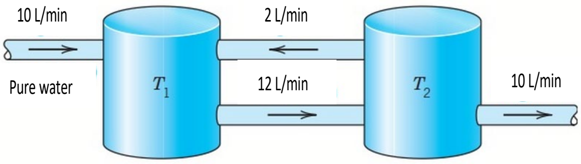
d. $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is an eigenvector of **A**

e. **A** is a singular matrix

14. A mass-spring system is given by the ODE: my” + cy’ + ky = 0
    where k is the spring constant, c is the damping constant, and m is the mass of the ball. The ODE can be converted to an SDE $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -k/m & -c/m \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$. Suppose the equilibrium point of the SDE is a degenerate node. Which of the following statements are correct? (Pick all correct answers)
    a. The equilibrium point is unstable.
    b. The ball will come to a halt as slow as possible without oscillation.
    c. c/2 = $\sqrt{mk}$
    d. The damping force in the system is equal to the spring force.
    e. None of the above

Degenerate node corresponds to critical damping, where ball will come to a halt as quickly as possible without oscillation, and the condition is given by c^2 = 4mk, which implies c/2 = $\sqrt{mk}$. Since the ball come to a halt, the equilibrium point is asymptotically stable. In general c and k are not equal.

15. Consider the two tanks in the diagram. Tank 1 holds 100 litre of salt water and tank 2 holds 50 litre of salt water.



Suppose the amount of salt in tank 1 and 2 are $y_1(t)$ and $y_2(t)$ respectively. Then the SDE that describe the amount of salt in the two tanks are given by $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \frac{1}{100} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ where a = _-12_, b = _4_, c = _12_, d = _-24_ (FIB).

Given the rate of salt in a tank y’ = inflow rate – outflow rate, we set up the two differential equations as follow:

$$y_1' = \frac{2}{50}y_2 - \frac{12}{100}y_1 = -\frac{12}{100}y_1 + \frac{4}{100}y_2$$

$$y_2' = \frac{12}{100}y_1 - \frac{2}{50}y_2 - \frac{10}{50}y_2 = \frac{12}{100}y_1 - \frac{24}{100}y_2$$

So $\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \frac{1}{100} \begin{pmatrix} -12 & 4 \\ 12 & -24 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

Long Questions (4 questions; total 45 marks)

Q16 (10 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q16 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Given three planes P_1, P_2, P_3 on the XYZ-space with equations x + y — z = 0, 2x – y + z = 0 and x + 2y - 2z = 0 respectively.

(i) [3 marks] Set up a 3 x 3 homogeneous system with the three equations and solve the system.

Ans:

$$\begin{array}{ccccccc} x & + & y & - & z & = & 0 \\ 2x & - & y & + & z & = & 0 \\ x & + & 2y & - & 2z & = & 0 \end{array}$$

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 2 & -1 & 1 & 0 \\ 1 & 2 & -2 & 0 \end{array}\right) \xrightarrow{GJE} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

Set z = t (parameter. Then first row gives x = 0 and second row gives y – t = 0 ⇒ y = t.

General solution: x = 0, y = t, z = t

(ii) [3 marks] Describe the intersection of the three planes <u>geometrically</u> and express this intersection in terms of a linear span.

Ans:

Since there is only one parameter in the general solution, the intersection is a line (that passes through the origin).

Since the general solution is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ t \\ t \end{pmatrix} = t\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$, this intersection can be expressed as span{ (0, 1, 1) }.

(iii) [4 marks] The planes with equations x + y - z = a, 2x – y + z = b and x + 2y - 2z = c are parallel to P_1, P_2, P_3 respectively, where a, b, c are some numbers, not all zero. Write down an expression of c in terms of a and b so that the three new planes have common intersect. Elaborate how you derive your answer.

Ans:

$$\left(\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 2 & -1 & 1 & b \\ 1 & 2 & -2 & c \end{array}\right) \xrightarrow{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & -3 & 3 & b-2a \\ 1 & 2 & -2 & c \end{array}\right) \xrightarrow{R_3-R_1} \left(\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & -3 & 3 & b-2a \\ 0 & 1 & -1 & c-a \end{array}\right)$$

$$\xrightarrow{R_3+\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & a \\ 0 & -3 & 3 & b-2a \\ 0 & 0 & 0 & c-a+\frac{1}{3}(b-2a) \end{array}\right)$$

For the three planes to have common intersection, the system must be consistent.

So $c - a + \frac{1}{3}(b-2a) = 0 \Rightarrow c = \frac{1}{3}(5a-b)$.

Q17 (10 marks)

<u>Answer this question by writing in the exam answer booklet</u>. Do not type in Examplify directly. Indicate Q17 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Two quantities X and Y are related by a quadratic equation:

$$X^2 + aX + b = Y \quad (*)$$

for some a and b. Four sets of measurements for (X, Y) are observed:

(2, 10), (3, 16), (4, 25), (5, 36)

(i) [2 marks] Use the measurements to set up a linear system **Ax** = **B** in variables a and b.

Ans:

$$\begin{array}{cccccc} 2a & + & b & = & 6 \\ 3a & + & b & = & 7 \\ 4a & + & b & = & 9 \\ 5a & + & b & = & 11 \end{array}$$

(ii) [5 marks] Find the least squares solutions of the above system, and write down the quadratic equation (*) that best fits the given data. Show your workings clearly. (You may use MATLAB to compute the standard matrix operations.)

Ans:

With A = $\begin{pmatrix} 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 6 \\ 7 \\ 9 \\ 11 \end{pmatrix}$, solve $A^TAx = A^TB$:

$$\Rightarrow \begin{pmatrix} 54 & 14 \\ 14 & 4 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 124 \\ 33 \end{pmatrix} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 54 & 14 \\ 14 & 4 \end{pmatrix}^{-1}\begin{pmatrix} 124 \\ 33 \end{pmatrix} = \begin{pmatrix} 1.7 \\ 2.3 \end{pmatrix}.$$

Least squares solution: a = 1.7, b = 2.3

Quadratic equation: $X^2 + 1.7X + 2.3 = Y$

(iii) [3 marks] For the matrix **A** and column vector **B** in (i), find the shortest possible distance between **B** and a vector in the column space of **A**. Give your answer up to 4 decimal places. Explain how you derive your answer.

Ans:

Any vector in the column space of A has the form Av for some 2-vector v in $\mathbf{R}^2$.

By definition of least squares solution of Ax = B, among all Av, the distance is the shortest between Ax_0 and B, where $x_0 = \begin{pmatrix} 1.7 \\ 2.3 \end{pmatrix}$ is the least squares solution in (ii).

So the shortest distance is given by $\|Ax_0 - B\| = \left\| \begin{pmatrix} 2 & 1 \\ 3 & 1 \\ 4 & 1 \\ 5 & 1 \end{pmatrix}\begin{pmatrix} 1.7 \\ 2.3 \end{pmatrix} - \begin{pmatrix} 6 \\ 7 \\ 9 \\ 11 \end{pmatrix} \right\| = 0.5477$

Q18 (12 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q18 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

The annual migration of populations between town A and town B are given by the following model: Each year, 15% of town A population moves to town B, and 10% of town B population moves to town A.

(i) [3 marks] Let a_n and b_n denote the populations in town A and B respectively at the end of year n. Find the matrix **A** such that $\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \mathbf{A}\begin{pmatrix} a_n \\ b_n \end{pmatrix}$. Show how your answer is derived.

Ans:

$$\begin{array}{rcccl} a_{n+1} & = & 0.85a_n & + & 0.1b_n \\ b_{n+1} & = & 0.15a_n & + & 0.9b_n \end{array} \Longrightarrow \begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \begin{pmatrix} 0.85 & 0.1 \\ 0.15 & 0.9 \end{pmatrix}\begin{pmatrix} a_n \\ b_n \end{pmatrix}$$

So A = $\begin{pmatrix} 0.85 & 0.1 \\ 0.15 & 0.9 \end{pmatrix}$

(ii) [2 marks] Express the n-th power $\mathbf{A}^n$ of **A** as a product $\mathbf{BCB}^{-1}$ where **B** is a non-singular matrix and **C** is a diagonal matrix.

Ans:

Eigenvalues of A are ¾ and 1, with corresponding eigenvectors $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2/3 \\ 1 \end{pmatrix}$.

So A = $\begin{pmatrix} -1 & 2/3 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \frac{3}{4} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 2/3 \\ 1 & 1 \end{pmatrix}^{-1}$ ⇒ A^n = $\begin{pmatrix} -1 & 2/3 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \left(\frac{3}{4}\right)^n & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & 2/3 \\ 1 & 1 \end{pmatrix}^{-1}$

(iii) [4 marks] Suppose the initial populations a_0 and b_0 in both towns are 100,000. Express each of a_n and b_n as an equation in terms of n. Show your workings clearly. (You may use MATLAB where possible.)

Ans: $\begin{pmatrix} a_n \\ b_n \end{pmatrix} = \begin{pmatrix} -1 & \frac{2}{3} \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \left(\frac{3}{4}\right)^n & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & \frac{2}{3} \\ 1 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 100{,}000 \\ 100{,}000 \end{pmatrix}$

$$= \begin{pmatrix} -1 & \frac{2}{3} \\ 1 & 1 \end{pmatrix}\begin{pmatrix} \left(\frac{3}{4}\right)^n & 0 \\ 0 & 1 \end{pmatrix}\frac{1}{5}\begin{pmatrix} -3 & 2 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 100{,}000 \\ 100{,}000 \end{pmatrix}$$

$$= 20{,}000\begin{pmatrix} -\left(\frac{3}{4}\right)^n & \frac{2}{3} \\ \left(\frac{3}{4}\right)^n & 1 \end{pmatrix}\begin{pmatrix} -1 \\ 6 \end{pmatrix} = 20{,}000\begin{pmatrix} \left(\frac{3}{4}\right)^n + 4 \\ -\left(\frac{3}{4}\right)^n + 6 \end{pmatrix}$$

a_n =20,000($\left(\frac{3}{4}\right)^n + 4$) and b_n =20,000($-\left(\frac{3}{4}\right)^n + 6$)

(iv) [3 marks] Explain why the ratio of the populations of the two towns in the long run is always the same regardless of the initial populations a_0 and b_0.

Ans: In the long run, A^n = $\begin{pmatrix} -1 & \frac{2}{3} \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & \frac{2}{3} \\ 1 & 1 \end{pmatrix}^{-1} = \frac{1}{5}\begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}$

For any initial populations a_0 and b_0, in the long run:

$$\begin{pmatrix} a_n \\ b_n \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} a_0 \\ b_0 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 2(a_0 + b_0) \\ 3(a_0 + b_0) \end{pmatrix}$$

So the ratio is always 2: 3.

Q19 (13 marks)

Answer this question by writing in the exam answer booklet. Do not type in Examplify directly. Indicate Q19 at the top on each page of the answer to this question. Indicate the part numbers on the left margin. You may use MATLAB for this question. But you should show how your answers are derived.

Consider the SDE $\begin{pmatrix} y_1{}' \\ y_2{}' \end{pmatrix} = \begin{pmatrix} -1 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$.

(i) [2 marks] Write down a fundamental set of solutions of the SDE.

Ans:

Eigenvalues of A are 3 and -3, with corresponding eigenvectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \end{pmatrix}$.

So a fundamental set of solutions is $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t}, \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t} \right\}$

(ii) [3 marks] Solve the initial value problem with $y_1(0) = 1$ and $y_2(0) = 0$. Show your working clearly.

Ans: General solution of the SDE is given by

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t}$$

With the given initial conditions,

$\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} \Rightarrow c_1 = 1/3$ and $c_2 = -1/3$.

So the particular solution of the IVP is

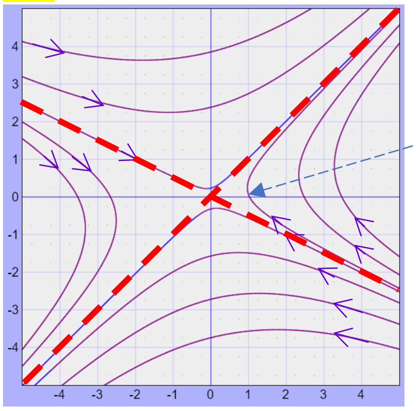
$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \end{pmatrix} e^{3t} + \begin{pmatrix} 2/3 \\ -1/3 \end{pmatrix} e^{-3t}$$

(iii) [2 marks] What is the type and stability of the equilibrium point of the SDE?

Ans: Since the two eigenvalues are of opposite sign, the equilibrium point is a saddle point, and is unstable.

(iv) [4 marks] Sketch a phase portrait of the SDE. Indicate in your diagram the trajectory that corresponds to the solution in (ii).

Ans:



Trajectory corresponds to solution in (ii)

(v) [2 marks] Write down a particular solution of the SDE that corresponds to a trajectory which moves closer and closer to the equilibrium point.

Ans: $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = k \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t}$ for any non-zero k.